

Function:

Definition: A function f from a set A to a set B , $f: A \rightarrow B$, is a rule of correspondence that assigns to each element x in the set A exactly one element y in the set B .

The set A is called the domain of the function f .

The range or codomain of the function is the set of elements in B that are in correspondence with elements in A .

In the case of functions, the variable x is the independent variable and the variable y is the dependent variable.

- x is the domain and $f(x)$ is the range value y for a given x .

- The process of finding the value of $f(x)$ for a given value of x is called evaluating a function.

Example: Let $f(x) = x^2$. Find the domain and range of $f(x)$.

For finding domain and range, i.e., when we plot the function, one question arises - what is graph of function or when we can say graph of function?

The graph of the function f consists of those points (x, y) such that $y = f(x)$. Not every curve is the graph of a function.

The reason is that a function assigns to a given input a single number as the output. If some line parallel to the y -axis meets the curve more than ~~once~~ once, then the curve is NOT the graph of a function.

Zoo of function

In mathematics there are many kinds of functions. Here is some of them.

- Polynomial functions :

1. linear : e.g., $f(x) = 2x - 1$

2. quadratic : e.g., $f(x) = -x^2$

3. Cubic : e.g., $f(x) = 4x^3 - 3x^2 + 5$

- Rational functions. e.g., $f(x) = \frac{2}{x-5}$

- Irrational functions. e.g., $f(x) = \sqrt{2-x}$

- Absolute value functions e.g., $f(x) = |x-3|$

- Exponential functions. e.g., $f(x) = 2^x$

- Logarithmic function e.g., $f(x) = \log_2(x)$, $f(x) = \ln(x)$

- Trigonometric function. e.g., $f(x) = \sin(x)$ $f(x) = \tan(x)$

- Even function. if $f(-x) = f(x) \quad \forall x \in \mathbb{R}$

- Odd function if $f(-x) = -f(x) \quad \forall x \in \mathbb{R}$

Transformations of functions

Let $f(x)$ be a function.

1. $y = f(x) + k$

- If $k > 0$ it moves up.

- If $k < 0$ it moves down.

2. $y = f(x + k)$

- If $k > 0$ it moves left

- If $k < 0$ it moves right

3. $y = k f(x)$

- If $k > 1$ stretches in the y -direction

- If $0 < k < 1$ compresses it

4. $y = f(kx)$

- If $k > 1$ compresses it in the x -direction

- If $0 < k < 1$ stretches it

5. $y = -f(x)$ reflects it about x -axis

6. $y = f(-x)$ reflects it about y -axis.

Composition of Two functions

The composition of the functions f and g is given by

$$(f \circ g)(x) = f(g(x))$$

The domain of the composite function, $f \circ g$, is the set of all x in the domain of g such that $g(x)$ is in the domain of f .

e.g. 1. Let $f(x) = 1 + 2x$ and $g(x) = x^2$

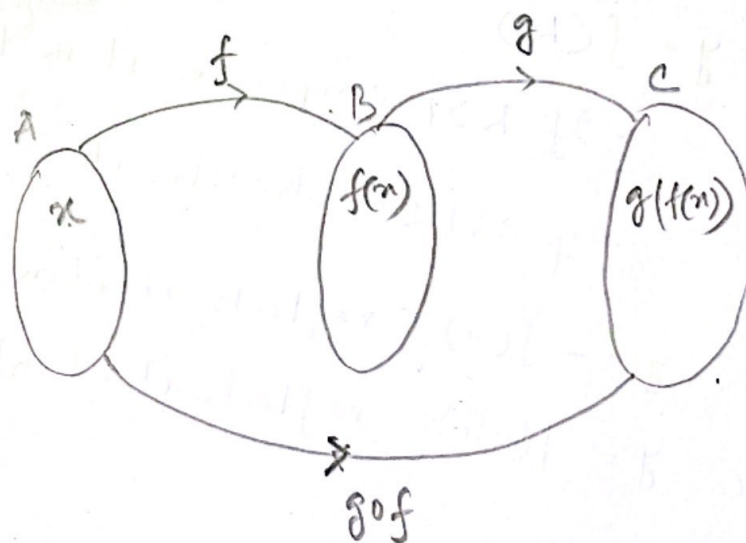
Compute $(f \circ g)(x)$ and $(g \circ f)(x)$.

2. Let $f(x) = \sqrt{x}$, $g(x) = x^3 - 1$.

Compute $(f \circ g)(x)$ and $(g \circ f)(x)$.

3. Let $f(x) = \sqrt{x}$, $g(x) = x^2$

Compute $(f \circ g)(x)$ and $(g \circ f)(x)$.



If $f: A \rightarrow A$ then $f^n := \underbrace{f \circ f \circ f \circ \dots \circ f}_{n \text{ times}} \quad \forall n \in \mathbb{N}$

e.g. $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $x \mapsto \sin(x)$ then $f \circ f: \mathbb{R} \rightarrow \mathbb{R}$ s.t. $x \mapsto \sin(\sin(x)) \quad \forall x \in \mathbb{R}$

Injective, Surjective, bijective and Inverse Functions

A function $f: X \rightarrow Y$ that assigns distinct outputs to distinct inputs is called an injective or one-to-one function. Hence, a function is injective if for every $a, b \in X$ such that $f(a) = f(b)$, then $a = b$.

The graph of a one-to-one function has the property that every horizontal line meets it in at most one point and if each horizontal line meets the graph of a function in at most one point, then the function is one-to-one.

The function is surjective or onto if every element of the codomain is mapped to by at least one element of the domain.

Hence, a function $f(x): X \rightarrow Y$ is surjective if the range of f is Y .

A function is bijective if it is BOTH injective and surjective.

Monotonic Functions:

If $f(x_1) < f(x_2)$ whenever $x_1 < x_2$, then $f(x)$ is an increasing function.

If $f(x_1) > f(x_2)$ whenever $x_1 < x_2$, then $f(x)$ is a decreasing function.

These TWO TYPES of functions are called monotonic.

Inverse Functions

Let $f(x)$ be a one-to-one function. The function $g(x)$ that assigns to each output of f the corresponding unique input is called the inverse of f . The symbol f^{-1} denotes the inverse function.

Example: Determine the inverse of the following functions and then graph them.

1. $f(x) = 2x$

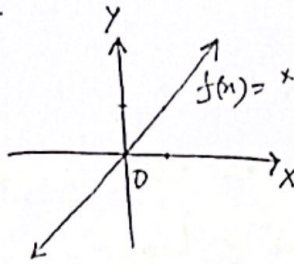
2. $f(x) = x^3$

3. $f(x) = x^2$

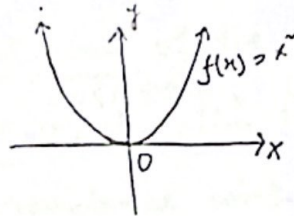
4. $f(x) = \frac{1}{x}$

Some Basic Functions

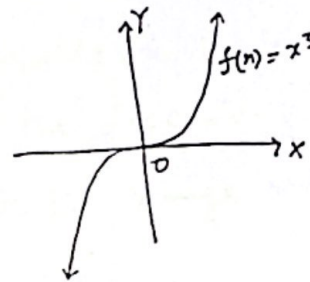
① $f(x) = x$



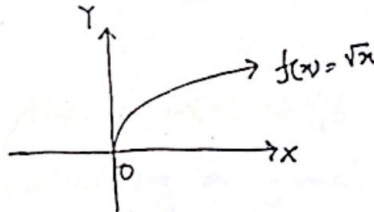
② $f(x) = x^2$



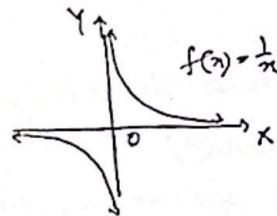
③ $f(x) = x^3$



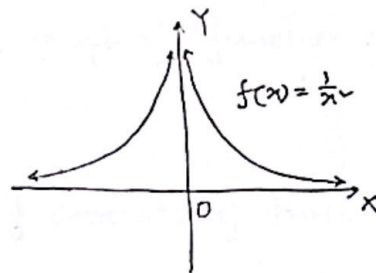
④ $f(x) = \sqrt{x}$, $x \geq 0$



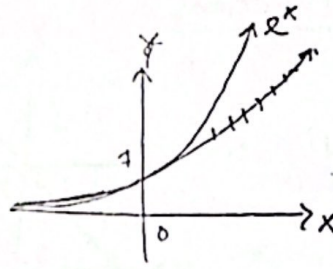
⑤ $f(x) = \frac{1}{x}$, $x \neq 0$



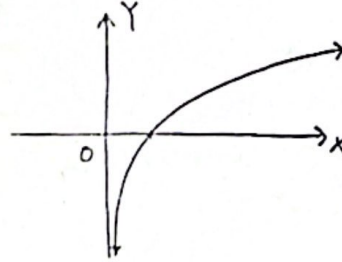
⑥ $f(x) = \frac{1}{x^2}$, $x \neq 0$



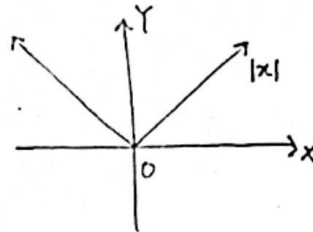
⑦ $f(x) = e^x$



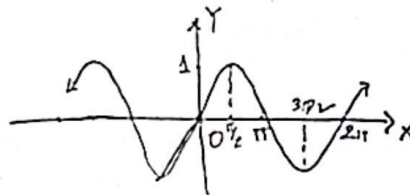
⑧ $f(x) = \ln(x)$, $x > 0$



⑨ $f(x) = |x|$



⑩ $f(x) = \sin(x)$



⑪ $f(x) = \cos(x)$

